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BAHÇEŞEHIR UNIVERSITY

FACULTY OF ENGINEERING AND NATURAL SCIENCES

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

PROJECT REPORT

AUDIO BALANCE INDICATOR

PROJECT NUMBER: 8

EEE3116 – ELECTRONICS 2
COURSE PROJECT

Ayşe Ece NAİI
Andaç ÜNAI

TABLE OF CONETENTS

COMPONENTS

1. SEMICONDUCTORS & INTEGRATED CIRCUITS (IC's)

- 1* LM741 Op-amp IC: The 8-pin black chip that serves as the brain of the circuit.
- 1* Red LED: The right-side indicator light.
- 1* Green LED: The left-side indicator light.
- 2* OA-91 Diodes: Used for rectifying the audio signal.

2. CAPACITORS

- 4* 10uF Electrolytic Capacitors: Couples the audio signal into the circuit and the filter.
- 1* 1uF Electrolytic Capacitor: Used to stabilize the virtual ground and the filter.

3. RESISTORS

- 4* 10k ohm resistors: Input resistors.
- 2* 100k ohm resistor: The feedback resistor connects pin 2 to 6.
- 1* 470 ohm resistor: The current-limiting resistor coming out of pin 6 and going into the middle junction of the LEDs.

4. POWER

- 1* 9V BATTERY

PROJECT OBJECTIVES

The primary objectives of this project are:

- **To design and construct** a visual Audio Balance Indicator using an operational amplifier (Op-Amp).
- **To process analog audio signals** from a dual-channel source (Left and Right channels) and convert them into a real-time visual representation.
- **To implement a Virtual Ground circuit** using a resistive voltage divider network, allowing a single-supply 9V battery to mimic a dual-rail +4.5V / -4.5V power supply required for symmetrical AC signal processing.
- **To demonstrate the differential amplification capabilities** of the LM741 operational amplifier in comparing two distinct audio input levels.

WHAT WE BUILT

We constructed a solid-state audio visualization circuit on a full-size breadboard. The circuit takes an audio input through a modified multi-core Aux cable, filters the signal through coupling capacitors, and processes it via an **LM741 Operational Amplifier IC**.

The output stage consists of a **bi-directional (anti-parallel) LED display** featuring a Red LED and a Green LED. These LEDs act as a visual bridge, dynamically reacting to the potential difference created at the output of the Op-Amp relative to the established virtual ground.

HOW IT WORKS

A. Power Supply & Virtual Ground Creation

Operational amplifiers traditionally require a dual-rail power supply (+Vcc and -Vcc) to process alternating current (AC) audio signals without clipping the negative cycles. To achieve this using a standard 9V battery, we implemented a voltage divider network consisting of two 100k ohm resistors.

- This configuration splits the 9V potential exactly in half, establishing a stable **+4.5V reference point**, known as the **Virtual Ground**.
- A 10uF capacitor is placed across this junction to filter out DC ripples and stabilize the reference voltage during heavy LED transitions.

B. Input Stage & Signal Coupling

The audio signal contains both DC offsets and AC audio waves. The 10uF electrolytic capacitors act as **DC-blocking (coupling) capacitors**, allowing only the pure AC audio frequencies to enter the system. The signals are then fed into the inverting (Pin 2) and non-inverting (Pin 3) inputs of the LM741 through 10k ohm protection resistors.

C. Operational Amplifier & Feedback Loop

The LM741 configured in a differential mode constantly compares the voltage levels between Pin 2 and Pin 3:

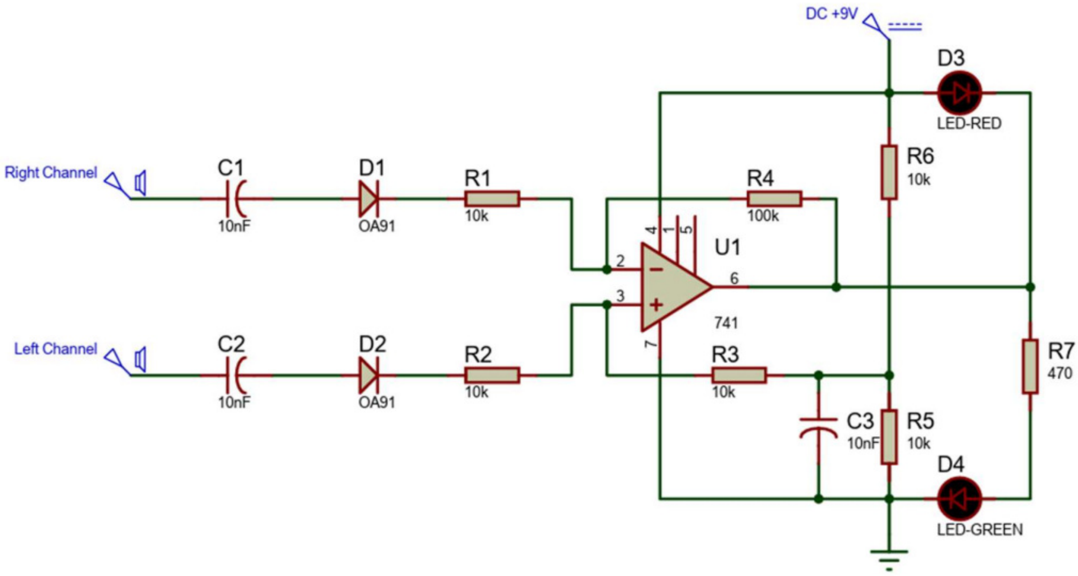
- A 100k ohm feedback resistor is securely connected between the inverting input (Pin 2) and the output (Pin 6). This sets the **closed-loop gain** (Gain ≈ R3 / R1), magnifying the tiny millivolt fluctuations coming from the Aux jack.
- When the Left channel signal amplitude exceeds the Right channel, the Op-Amp output swings towards the negative rail (0V).
- Conversely, when the Right channel dominates, the output swings towards the positive rail (+9V).

D. Visual Output Stage

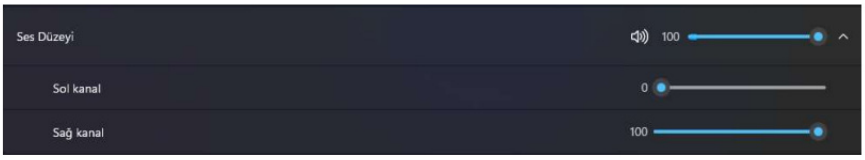
The Red and Green LEDs are connected in an **anti-parallel** layout. Their shared junction is tied to the Virtual Ground +4.5v, while the other side is connected to the Op-Amp output (Pin 6) via 470-ohm current-limiting resistor.

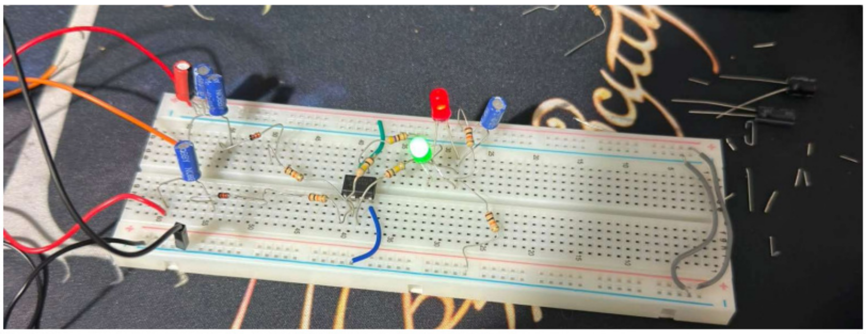
- If Pin 6 goes higher than +4.5V, current flows in one direction, illuminating the **Red LED**.
- If Pin 6 drops below +4.5V, current reverses direction, illuminating the **Green LED**.
- When the audio signal is perfectly balanced or silent, the output stays at exactly +4.5V (matching the virtual ground), keeping both LEDs turned off or dimly balanced.

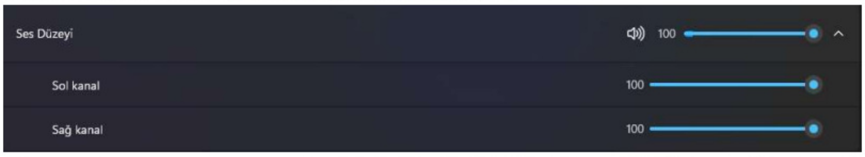
SCHEMATIZED VERSION ON PROTEUS

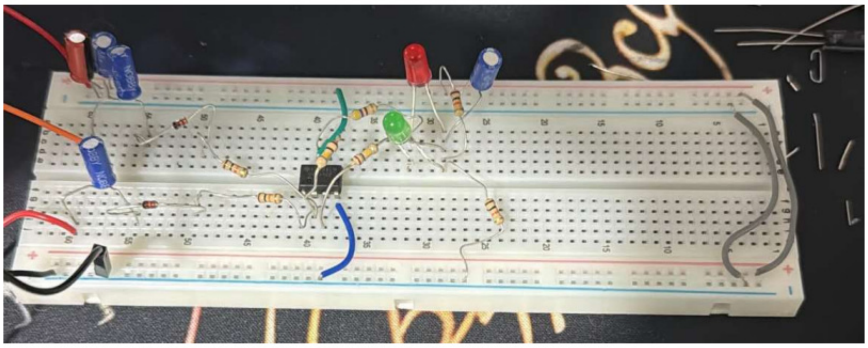


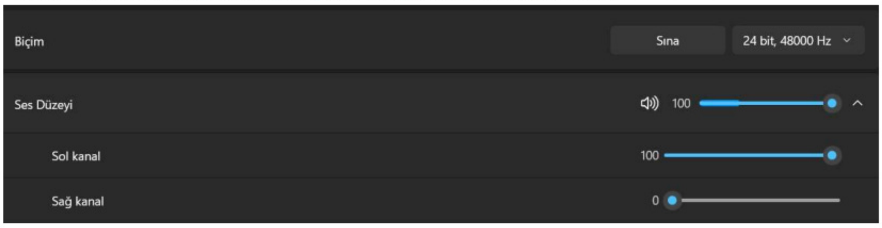
PHOTOS FROM PROJECT

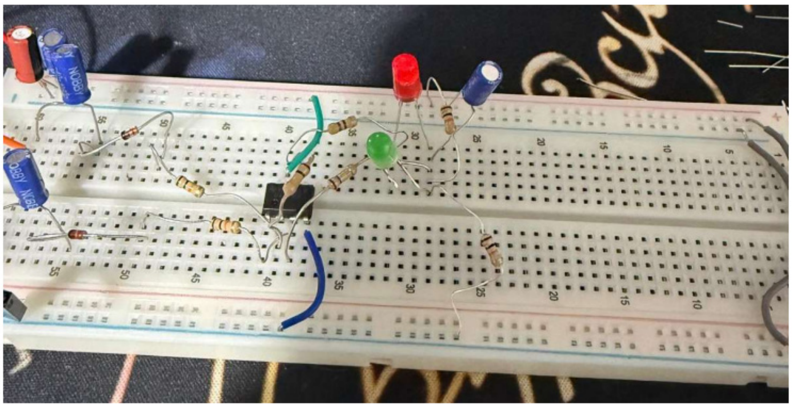












CONCLUSION

The design, assembly, and testing of the Audio Balance Indicator were successfully completed. Through a systematic troubleshooting process, several critical concepts in practical analog electronics were validated:

- **The Importance of Power Reference:** Implementing a Virtual Ground network successfully allowed a single 9V battery to power an AC-coupled operational amplifier circuit, proving that dual-rail supply requirements can be met efficiently through passive component configuration (R5, R6, and C3).
- **Signal Integrity and Connectivity:** The project highlighted the practical challenges of dealing with multi-core audio cables, specifically the presence of insulating make coatings on headphone wires and the absolute necessity of a shared common ground (GND) between the signal source and the processing circuit to allow proper voltage reference.
- **Component Optimization:** Bypassing the high forward-voltage threshold of standard silicon diodes allowed the circuit to maintain high sensitivity, ensuring that low-voltage audio signals from commercial headphone jacks are strong enough to drive the LM741 differential stage.

In conclusion, the circuit functions reliably as a real-time audio visualizer. When a balanced audio signal is introduced, the anti-parallel LED output dynamically reflects the stereo shifts between the left and right channels, fulfilling all the initial engineering and educational objectives of the project.